

Module 1: Hardware and Components Basics

Understanding of Hardware and its components

DAY 5:-

Task 1:

The following components inside the Desktop Pc are:

1. **Motherboard** - The main circuit board that connects and allows communication between all computer components.
2. **CPU** - The brain of the computer that performs calculations and executes instructions.
3. **RAM** - Temporary high-speed memory used to store data and programs currently in use.
4. **GPU** - Processes graphics, videos and games, improving visual performance.
5. **Power Supply Unit** - Converts AC power from the wall into DC power for the computer components.
6. **Storage Drives** - Stores the operating system, applications, games, and personal files permanently.
7. **Case Fans** - Improve airflow inside the cabinet.
8. **SATA Cables** - Connect storage devices to the motherboard for data transfer.
9. **BIOS/UEFI Chip** - Firmware chip that initializes hardware and starts the operating system during boot.
10. **Expansion Cards** - Additional cards (Wi-Fi, sound, capture cards, etc.) inserted into motherboard slots for extra functionality.

Task 2:

Desktop PC Disassembly Steps

Step 1- Shut Down and Unplug the PC

Turned off the desktop PC completely and unplugged the power cable from the power supply to avoid electrical hazards before starting disassembly.

Step 2 - Remove External Connections

Disconnected all external devices and cables such as:

- keyboard
- mouse
- monitor
- Ethernet cable
- USB devices
- speakers
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Step 3 - Open the Side Panel

Removed the screws from the side panel and carefully slid the panel off to access internal components of the PC.



Step 4 - Observe Internal Components

Identified major components inside the cabinet:

- Motherboard
- CPU cooler
- RAM
- GPU
- HDD/SSD
- power supply

Step 5 - Disconnect Power Supply Cables

Disconnected:

- 24-pin motherboard power connector
- CPU power connector
- SATA power cables

- Front Panel Header

Step 6 - Remove the RAM

Unlocked the RAM slot clips and carefully removed the RAM sticks by holding them from the edges to avoid damaging the contacts.



Step 7 - Remove HDD / SSD Storage Drives

Disconnected SATA data and power cables from the HDD/SSD.

Step 8 - Remove the GPU

Unscrewed the graphics card bracket, released the PCIe slot lock, and gently removed the GPU from the motherboard.



Step 9 - Remove Cooling Fans

Disconnected fan headers and removed the CPU cooler and case fans to expose motherboard components.

Step 10 - Remove the Motherboard

Unscrewed all motherboard mounting screws and carefully lifted the motherboard out of the cabinet while ensuring no cables remained connected.

Step 11 - Remove the Power Supply Unit (PSU)

After disconnecting all power cables and removing major components, unscrewed the PSU from the rear of the cabinet and carefully removed it from the case.



Task 3:

Desktop PC Reassembly Steps

Step 1 - Prepare the Cabinet and Components

Placed the cabinet on a stable surface and organized all removed components and screws before starting reassembly. This helps avoid missing parts and improves efficiency.

Step 2 - Install the Power Supply Unit (PSU)

Inserted the PSU into its mounting position and secured it with screws at the rear of the cabinet.

Why the order matters:

The PSU is installed early because it powers all components and its cables need to be routed properly before other hardware blocks access.

Step 3 — Install the Motherboard

Carefully aligned the motherboard with the cabinet standoffs and I/O shield, then secured it using motherboard screws.

Why the order matters:

The motherboard acts as the main platform for connecting all other components.

Step 4 - Connect Motherboard Power Cables

Connected:

- 24-pin ATX power connector
- CPU power connector

from the PSU to the motherboard.

Why the order matters:

These are the primary power connections required before powering the system.



Step 5 - Install the CPU Fans

Mounted the CPU cooler and connected the fan cable to the CPU FAN header on the motherboard.

Why the order matters:

The CPU must have cooling before the system is powered on to prevent overheating.

**Step 6 - Install the RAM**

Aligned the RAM stick with the RAM slots and pressed them in until the side clips locked.

Why the order matters:

RAM must be installed before booting because the system cannot operate without memory.

**Step 7 - Install HDD / SSD Storage Drives**

Mounted the HDD/SSD into the drive bay and connected:

- SATA data cable to motherboard
- SATA power cable from PSU

Why the order matters:

Storage devices are required for the operating system and file storage.



Step 8 - Install the GPU

Inserted the graphics card into the PCIe x16 slot and secured it with screws.

Why the order matters:

The GPU is installed after the motherboard and PSU cables to provide easier access to slots and connectors.



Step 9 - Connect Front Panel and Internal Cables

Connected:

- power switch
- reset switch
- front USB headers
- front audio headers

Why the order matters:

These small connectors are easier to attach after major components are installed.

Step 11 - Close the Side Panel

Reattached the side panel and tightened the screws securely.

Step 12 - Connect External Devices

Connected:

- monitor
- keyboard
- mouse
- power cable
- Ethernet cable

Step 13 - Power On and Test the System

Turned on the PC and verified:

- fans spinning,
- display output,
- BIOS/OS boot,
- proper hardware detection.

This confirms successful reassembly of the desktop PC.



Task 4:

Safety Precautions Checklist:

Safety Precautions	Why it is important
Shut down the PC properly	Prevents data corruption and hardware damage.
Unplug the power cable	Eliminates risk of electric shock and short circuits.
Switch off PSU power button	Ensures no standby power remains in the system.
Work on a clean and flat surface	Prevents components from falling or getting damaged.
Wear an anti-static wrist strap	Protects sensitive components from static electricity.
Handle components by the edges	Prevents damage to circuits and connector pins.
Keep liquids away from workspace	Avoids short circuits and accidental damage.
Organize screws separately	Prevents losing screws and mixing component hardware.
Avoid forcing connectors or parts	Prevents breaking ports, slots, and cables.
Ensure proper lighting	Makes connectors and screws easier to see safely.

Tools Required Checklist:

Tool	Why it is important
Screwdriver	Used to remove and install most PC screws.
Small container or magnetic tray	Keeps screws and small parts organized safely.
Cleaning cloth	Helps clean dust and components safely.
Compressed air blower	Removes dust from fans, motherboard, and heatsinks.
Flashlight	Helps view connectors and motherboard headers clearly.
Plastic pry tool	Helps open tight panels without scratching components.

